

23.4

Q2: Rationalize the denominators of the following:

i. $\frac{1}{\sqrt{2} + 10}$

so

$$\frac{1}{\sqrt{2} + 10} \times \frac{\sqrt{2} - 10}{\sqrt{2} - 10}$$

Note: Rationalize means multiply and divide the conjugate of Denominator.

$$\frac{\sqrt{2} - 10}{(\sqrt{2})^2 - (10)^2} = \frac{\sqrt{2} - 10}{2 - 100}$$

$$\frac{\sqrt{2} - 10}{-98}$$

$$\boxed{\frac{-\sqrt{2} + 10}{98}}$$

Ans

ii. $\frac{1}{3 + 2\sqrt{5}} \times \frac{3 - 2\sqrt{5}}{3 - 2\sqrt{5}}$

$$\frac{3 - 2\sqrt{5}}{(3)^2 - (2\sqrt{5})^2} = \frac{3 - 2\sqrt{5}}{9 - 4 \times 5}$$

$$= \frac{3 - 2\sqrt{5}}{9 - 20} = \frac{3 - 2\sqrt{5}}{-11}$$

$$= \frac{-3 + 2\sqrt{5}}{11}$$

Ans

iii $\frac{1}{4\sqrt{3} - 3\sqrt{6}}$

Sol

$$\frac{1}{4\sqrt{3} - 3\sqrt{6}} \times \frac{4\sqrt{3} + 3\sqrt{6}}{4\sqrt{3} + 3\sqrt{6}}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{(4\sqrt{3})^2 - (3\sqrt{6})^2}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{(16 \times 3) - (9 \times 6)}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{48 - 54}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{12 - 56} = \frac{-4\sqrt{3} - 3\sqrt{6}}{56}$$

Ans

iii $\frac{1}{4\sqrt{3} - 3\sqrt{6}}$

sol

$$\frac{1}{4\sqrt{3} - 3\sqrt{6}} \times \frac{4\sqrt{3} + 3\sqrt{6}}{4\sqrt{3} + 3\sqrt{6}}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{(4\sqrt{3})^2 - (3\sqrt{6})^2}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{(16 \times 3) - (9 \times 6)}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{48 - 54}$$

$$\frac{4\sqrt{3} + 3\sqrt{6}}{12 - 54} = \frac{-4\sqrt{3} - 3\sqrt{6}}{54}$$

Ans